

Module-5

Data Analysis and Report Writing: Editing, Coding, Classification, Tabulation, Validation. Analysis and Interpretation, Report writing and presentation of results, Importance of report writing, types of research reports, Report structure, Guidelines for effective documentation.

Data Analysis and Report Writing: Editing

The data which has been collected from various primary/secondary sources is RAW in nature, this means that there are likely chances of errors and inconsistencies in it.

Since data collected is of pivotal importance to policy and decision makers everywhere be it governmental departments, business organizations, health or educational institutions etc it would be better to have a team of experts at hand who know how to scrutinize, review and **edit** this data before it is finally fed into the data bases and the required statistics are generated.

How to define data editing?

The process through which the data is reviewed to check for consistency, adequacy, detect errors and outliers (values that are either too big or too small from the rest of the data) and the correction of errors within the data in order to maximize its usefulness for the purpose for which it was collected is called **data editing**.

Purposes and objectives of data editing

The basic purpose served by data editing is that it improves the quality, accuracy and adequacy of the collected data thereby making it more suitable for the purpose for which the data was collected. The following can therefore be identified as the main objectives of the data editing process:

- Detection of errors in the data that otherwise affect the validity of outputs.
- Validation of data for the purposes it was collected.
- Provision of information that that would help assess the overall level of accuracy of the data.
- Detection and identification of any inconsistencies in the data and outliers and to make adjustments for them.

Types of data editing

There are different types of data editing

- **Validity and completeness of data:** refers to correctness and completeness of obtained responses. This helps ensure that there are no missing values or empty fields in the data bases.
- **Range:** verifies that data within a field fall between the boundaries specified for the particular field.

- **Duplicate data entry:** this helps ensure that there is no repetition or duplication of data and each unit on the data base or register was filled only once.
- **Logical consistency:** through this type of editing connections between data fields or variables are taken into account.
- **Outliers:** this type of editing helps detect values that are too extreme or unusual so that they can be verified and checked.

Stages of data editing

The data editing process can be divided into three parts or stages

- The first stage is the stage where you set the rules for editing. This stage is further subdivided into two steps.

In step one, you provide instructions to desk editors who then check the data for coherence and consistency.

In step two, you set the rules by establishing logical relations between the variables according to various criteria. This set of rules is called automated validation rules and this type of editing seeks to detect errors during data entry and to screen them.

- The **manual desk editing** stage is a traditional method that is put into effect by a specialized editing team. The data, (if) on paper is checked after the data has been collected and before it is fed into the data bases. If however, electronic means have been used to collect the data, the forms entered into the database are revised individually.
- The **automated data editing** method makes use of computer programs and systems for checking the data all at once after it has been entered electronically. These programs and systems contain **Audit rules** which validate the data, detect errors and determine unacceptable responses.

Limitations to data editing

There are certain factors that can influence or pose limitations to the data editing process, these can be summed up as follows;

- Data editing can be influenced by the amount of time available, the budget, the presence or absence of other resources and also by the group of people involved in the editing process.
- The available computer software programs.
- Follow up with the respondents is of critical importance in the data editing process because they are often the best source of information in many cases. However, the respondents might feel this to be stressful and burdensome thereby causing limitations to the data editing process.

- Some types of data do not require extensive editing, therefore it would be better to keep in mind the intended uses of data and make sure that the more important part of data is kept free from all errors. In this way, the intended use of data does play an important role in influencing the data editing process.
- What you need to do is to establish the methods and procedures that must be followed while correcting or handling the data errors, in the survey plan, right at the start of the project otherwise the process would be of no or little use.
- Also remember that if you plan to edit your data manually, you must develop and document the methods that are to be followed. Your team must be trained, a method must be established to check their work progress and the impact of the edits on the original data must also be assessed.
- In case of automated editing, you would need to develop and document the rules for editing. You might also need to develop a software or customize an existing computer program as per your data editing demands or requirements.

General guidelines for data editing

There are certain things that must be kept in mind when editing the data, they are as follows;

- Who should make or set the editing rules? The answer to this question would be that such rules should be made by professionals who are **experts** in data collection, questionnaire design and analysis.
- The editing rules need to be consistent and free from any contradictions.
- When setting the editing rules, it must be established whether the variable is qualitative or quantitative because the rules for editing either one are different from the other.
- Give enough time to each of the various stages of the process, that is, data collection entry and analysis and at the end of each make a quick check to see that all the necessary edits have been made and that there are no empty places within the questionnaire form.
- The questionnaire must be edited in full during the early stages of editing. If however it is found that some errors remain, a sample of forms should be subjected to re editing. The size of the sample is determined according to the expected number of the remaining errors.
- You also need to re run the desk editing stage to ensure that the data is almost free from all errors.
- The questionnaire must be subjected to desk editing and also to automated rules within the built in data editing/computer software programs.

Coding

Introduction

When a researcher has completed collecting information or data, this information is ready to be processed and analyzed. Quantitative data is information that is measurable and focuses on numerical values, unlike qualitative data which is more descriptive. During the data processing step, the collected data is transformed into a form that is appropriate to manipulate and analyze.

The process in which raw data is transformed into a standardized form suitable for machine processing and analysis is called coding. In other words, coding is the act of assigning numerical values to a set of data in order to make the analysis simpler. Coding can be used to quantify both manifest content i.e. the tangible or concrete surface content (data), and latent content i.e. the underlying meaning behind this information. The difference between manifest content and latent content is very important when it comes to survey research.

It is advisable to do a pilot or a pretest of the instrument of data collection as it would help uncover the potential problems with the study and accordingly help make changes in the tool. It will also give the researcher an idea of how the data will look. On the basis of this, the researcher can work out the layout of the codebook keeping in mind the responses collected for each variable, guiding him to provide enough variables to capture all the richness, complexity, and variety of data that has been collected. Depending on what shape the data comes in, the researcher will have to decide how to code this information, with the help of one, two, or multiple variables.

Data Coding

Though the preparation of a codebook commences prior to actual data collection, after designing the instrument and pre testing it, data coding as a step in the research process takes place after the completion of data collection, simultaneously during data entry. It is important to keep the following points in mind during coding of data:

- **Identification variables:** - Unique identification of the respondent/ questionnaire/ response sheet is extremely important as it helps the researcher verify and check data. The identification variable is a unique number corresponding to every respondent which has to be accommodated in a special field at the beginning of each record. For example, 001, 002, etc. may be used as identification variables
- **Code categories:-** Code categories should be mutually exclusive, exhaustive, and precisely defined. Each interview response should fit into one and only one category. Ambiguity will cause coding difficulties and problems with the interpretation of the data also. An example of this would be while recording literacy levels of youth in a slum community, the coding should include not just those who have gone through the formal system of education, but also those who have participated in non-formal education programs and are therefore literate.

- **Preserving original information:-** Data once coded is retained and becomes final; hence it is important to code as much detail as possible by recording the original data rather than collapsing or bracketing the information. With original or detailed data, the research analyst can determine other meaningful relationships between variables beyond those which are selected primarily or restricted by the entering or coding data. Hence, occupation of women in a slum community being surveyed could include the home based small enterprises (involved in small scale business like making snacks, Knick knacks, etc.) each as a separate code.
- **Closed-ended questions:-** Responses to survey questions that are pre coded in the questionnaire should retain this coding scheme in the machine-readable data to avoid errors and confusion. For example, in the above mentioned study in case the women respondents may be pre divided as per their age and marital status, this pre coding should be retained.
- **Open-ended questions:-** For open-ended items, investigators can either use a predetermined coding scheme or review the initial survey responses to construct a coding scheme based on major categories that emerge. Any coding scheme and its derivation should be reported in study documentation. Increasingly, investigators submit the full verbatim text of responses to open-ended questions to archives so that users can code these responses themselves. However, such responses may contain sensitive information and may involve the risk of identification; they must therefore be reviewed prior to disclosure.
- **Check-coding:-** Check-coding provides an important means of quality control in the coding process. In this process some cases are repeated with an independent coder in order to verify the coding assigned and rule out discrepancies and ambiguities if any.
- **Series of responses:-** If a series of responses requires more than one field, organizing the responses into meaningful major classifications becomes helpful. Responses within each major category are assigned the same first digit. Secondary digits can distinguish specific responses within the major categories. Such a coding scheme permits analysis of the data using broad groupings or more detailed categories.

Dealing with Missing Data

There are various situations wherein the data may be missing and each of these would need to be coded differently. Some of these situations are listed below:

- **Refusal to answer or No response** - In such a scenario the respondent explicitly refuses to answer a question or does not answer it when he or she should have. This may be with regard to questions that may be deemed personal and sensitive or too private by the respondent or even in a case where there was lack of clarity with regard to a particular question.
- **Don't know responses** - In this case the respondent was unable to answer a question, either because he or she had no opinion or because the required information was not available (e.g., a respondent could not provide family income for the previous year).

- **Processing error** - In this case, there is no answer to the question, although the subject provided one. This would indicate an error on the part of the interviewer, incorrect coding or other such problems, despite the respondent providing a response to the particular question.
- **Not applicable** - The subject was never asked a question for a particular reason. While this may be a result of skip patterns following filter questions, for example, subjects who are not married are not asked questions pertaining to their children. Other examples of inapplicability are sets of items asked only of random subsamples and those asked of one member of a household but not another.
- **No match** - This situation arises when data are drawn from different sources (for example, a survey questionnaire and an administrative database), and information from one source cannot be located
- **No data available** - The question should have been asked to the respondent, but for a reason other than those listed above, no answer was given or recorded. This may be due to an error on the part of the interviewer.

Classification,

Meaning of Classification of Data

- It is the process of arranging data into homogeneous (similar) groups according to their common characteristics.
- Raw data cannot be easily understood, and it is not fit for further analysis and interpretation. Arrangement of data helps users in comparison and analysis.
- For example, the population of a town can be grouped according to sex, age, marital status, etc.

Definition of classification given by Professor. Secrist - "Classification is the process of arranging data into sequences according to their common characteristics or separating them into different related parts."

Classification of data

The method of arranging data into homogeneous classes according to the common features present in the data is known as classification.

A planned data analysis system makes the fundamental data easy to find and recover. This can be of particular interest for legal discovery, risk management, and compliance. Written methods and sets of guidelines for data classification should determine what levels and measures the company will use to organise data and define the roles of employees within the business regarding input stewardship.

Once a data -classification scheme has been designed, the security standards that stipulate proper approaching practices for each division and the storage criteria that determines the data's lifecycle demands should be discussed.

Objectives of Data Classification

The primary objectives of data classification are:

- To consolidate the volume of data in such a way that similarities and differences can be quickly understood. Figures can consequently be ordered in sections with common traits.
- To aid comparison.
- To point out the important characteristics of the data at a flash.
- To give importance to the prominent data collected while separating the optional elements.
- To allow a statistical method of the materials gathered.

Tabulation

What is the meaning of Tabulation?

Tabulation is the systematic and logical representation of figures in rows and columns to ease comparison and statistical analysis. It eases comparison by bringing related information closer to each other and helps further in statistical research and interpretation. In other words, tabulation is a method of arranging or organizing data in a tabular form. The tabulation process may be simple or complex depending upon the type of categorization.

Tabulation Definition

Tabulation is defined as the process of placing classified data in tabular form. A table is a systematic arrangement of statistical information in rows and columns. The rows of a table are the horizontal arrangement of data whereas the columns of a table are the vertical arrangement of data.

What are the Essential Parts of a Table?

To tabulate data correctly, one must learn about the eight essential parts of a table. These are as follows –

1. Table Number –

This is the first part of a table and is given on top of any table to facilitate easy identification and for further reference.

2. Title of the Table –

One of the most important parts of any table is its title. The title is either placed just below the table number or at its right. It is imperative for the title to be brief, crisp and carefully-worded to describe the tables' contents effectively.

3. Headnote –

The headnote of a table is presented in the portion just below the title. It provides information about the unit of data in the table, like “amount in Rupees” or “quantity in kilograms”, etc.

4. Column Headings or Captions –

The headings of the columns are referred to as the caption. It consists of one or more column heads. A caption should be brief, short, and self-explanatory. Column heading is written in the middle of a column in small letters.

5. Row Headings or Stubs –

The title of each horizontal row is called a stub.

6. Body of a Table –

This is the portion that contains the numeric information collected from investigated facts. The data in the body is presented in rows which are read horizontally from left to right and in columns, read vertically from top to bottom.

7. Footnote –

Given at the bottom of a table above the source note, a footnote is used to state any fact that is not clear from the table's title, headings, caption or stub. For instance, if a table represents the profit earned by a company, a footnote can be used to state if said profit is earned before, or after tax calculations.

8. Source Note –

As its name suggests, a source note refers to the source from where the table's information has been collected.

Tabulation of Data: Illustration

Following is a tabulation of data to represent the total number of boys and girls in classes V, VI and VII in XYZ School.

Table Number – 1**Gender-Wise Distribution of Students in Classes- V, VI and VII of XYZ School
(Head Note)**

Gender	V	VI	VII	Total
Boys	50	60	65	175
Girls	45	50	60	155
Total	95	110	125	330

Types of Tabulation**1. Simple Tabulation or One-way Tabulation**

When the data in the table are tabulated to one characteristic, it is termed as a simple tabulation or one-way tabulation.

For example, Data tabulation of all the people of the World is classified according to one single characteristic like religion.

2. Double Tabulation or Two-way Tabulation

When the data in the table are tabulated considering two different characteristics at a time, then it is defined as a double tabulation or two-way tabulation.

For example, Data tabulation of all the people of the World is classified by two different characteristics like religion and sex.

3. Complex Tabulation

When the data in the table are tabulated according to many characteristics, it is referred to as a complex tabulation.

For example, Data tabulation of all the people of the World is classified by three or more characteristics like religion, sex, and literacy, etc.

What are the Objectives of Tabulation?

Tabulation essentially bridges the gap between the collection of data and analysing them. The primary objectives of tabulation are given below –

- **For Simplification of Complex Data –**

When any information is tabulated, the volume of raw data is compressed and presented in a much more simplified manner. This facilitates easy comprehension and analysis of previously complex data.

- **To Highlight Important Information –**

Representing any data in tabular form increases the scope to highlight important information. Since data is presented in a concise manner without any textual explanation, any crucial information is automatically highlighted without difficulty.

- **To Enable Easy Comparison –**

When data is presented in an orderly fashion in rows and columns, it becomes easier to compare between them on the basis of several parameters. For example, it becomes easier to determine the month when a country has received the maximum amount of rainfall if the data is presented in a table. Otherwise, there always remains room for making a mistake in processing the data correctly.

- **To Help in the Statistical Analysis of Data –**

Statistical analysis involves computing correlation, average, dispersion, etc. of data. When information is presented in an organised manner in a table, statistical analysis becomes a lot simpler.

- **Saves Space**

Even though it might not seem as important as the other objective of tabulation, saving space without sacrificing the quality of data can be extremely helpful in the long run. Additionally, a table helps to present facts in a much more concise manner than page after page of text.

How is Data in Tabulation Executed?

With tabulation, data can be carried out manually or with the help of a computer. Mostly, the execution of data tabulation depends on the cost, type, and size of the study, availability of computers, time at disposal, and such other factors.

If tabulation is carried out on a computer, the answers are converted in numeric form. Whereas in the case of tabulation by hand, one can utilize lists, tally, card sort, and count methods.

Following are the categories of tabulating data by hand:

- **Direct Tally Method –**

Here, codes are first written down in tally sheets. Then a stroke is marked against codes to denote response. After every fourth stroke, the fifth response is given by putting a horizontal or diagonal line through the stroke.

- **Card Sort and Count Method –**

This is perhaps the most efficient hand tabulation method. Here the data is recorded in cards of various sizes and shapes with the help of a series of holes. Next, cards belonging to each of the categories are segregated and counted, and their frequency is recorded. This way, a total of 40 items can be included in a single page.

- **List and Tally Method –**

With this method, a large number of questionnaires are listed in one sheet. The responses to each question are then entered into rows, and the code corresponding to each question is represented in columns.

What are the Rules of Tabulation?

There are a few general rules that have to be followed while constructing tables. These are –

- The tables illustrated should be self-explanatory, simple and attractive. There should be no need for further explanation (details). If the volume of information is substantial, it is best to put them down in multiple tables instead of a single one. This reduces the chances of mistakes and defeats the purpose of forming a table. However, each table formed should also be complete in itself and serve the purpose of analysis.
- The number of rows and columns should be kept minimal to present information in a crisp and concise manner.
- Before tabulating, data should be approximated, wherever necessary.
- Stubs and captions should be self-explanatory and should not require the help of footnotes to be comprehended.
- If certain positions of data collected cannot be tabulated under any stub or captions, they should be put down in a separate table under the heading `` miscellaneous.
- Quantity and quality of data should not be compromised under any scenario while forming a table.

Validation

What is Data Validation?

Data validation refers to the process of ensuring the accuracy and quality of data. It is implemented by building several checks into a system or report to ensure the logical consistency of input and stored data.

In automated systems, data is entered with minimal or no human supervision. Therefore, it is necessary to ensure that the data that enters the system is correct and meets the desired quality standards. The data will be of little use if it is not entered properly and can create bigger downstream reporting issues. Unstructured data, even if entered correctly, will incur related costs for cleaning, transforming, and storage.

Types of Data Validation

There are many types of data validation. Most data validation procedures will perform one or more of these checks to ensure that the data is correct before storing it in the database. Common types of data validation checks include:

1. Data Type Check

A data type check confirms that the data entered has the correct data type. For example, a field might only accept numeric data. If this is the case, then any data containing other characters such as letters or special symbols should be rejected by the system.

2. Code Check

A code check ensures that a field is selected from a valid list of values or follows certain formatting rules. For example, it is easier to verify that a postal code is valid by checking it against a list of valid codes. The same concept can be applied to other items such as country codes and NAICS industry codes.

3. Range Check

A range check will verify whether input data falls within a predefined range. For example, latitude and longitude are commonly used in geographic data. A latitude value should be between -90 and 90, while a longitude value must be between -180 and 180. Any values out of this range are invalid.

4. Format Check

Many data types follow a certain predefined format. A common use case is date columns that are stored in a fixed format like “YYYY-MM-DD” or “DD-MM-YYYY.” A data validation procedure that ensures dates are in the proper format helps maintain consistency across data and through time.

5. Consistency Check

A consistency check is a type of logical check that confirms the data’s been entered in a logically consistent way. An example is checking if the delivery date is after the shipping date for a parcel.

6. Uniqueness Check

Some data like IDs or e-mail addresses are unique by nature. A database should likely have unique entries on these fields. A uniqueness check ensures that an item is not entered multiple times into a database.

Practical Example

Consider the example of a retailer that collects data on its stores but fails to create a proper check on the postal code. The oversight could make it difficult to leverage the data for information and business intelligence. Several problems can occur if the postal code is not entered or entered improperly.

It can be difficult to define the location of the store in some mapping software. A store postal code will also help generate insights about the neighborhood where the store is located. Without a data check on the postal code, it is more likely to lose the value of data. It will result in further costs if the data needs to be recollected or the postal code needs to be manually entered.

A simple solution to the problem would be to put a check in place that ensures a valid postal code is entered. The solution could be a dropdown menu or an auto-complete form that allows the user to choose the postal code from a list of valid codes. Such a type of data validation is called a code validation or code check.

Data Validation in Excel

The following example is an introduction to data validation in Excel. The data validation button under the data tab provides the user with different types of data validation checks based on the data type in the cell. It also allows the user to define custom validation checks using Excel

formulas. The data validation can be found in the Data Tools section of the Data tab in the ribbon of Excel:

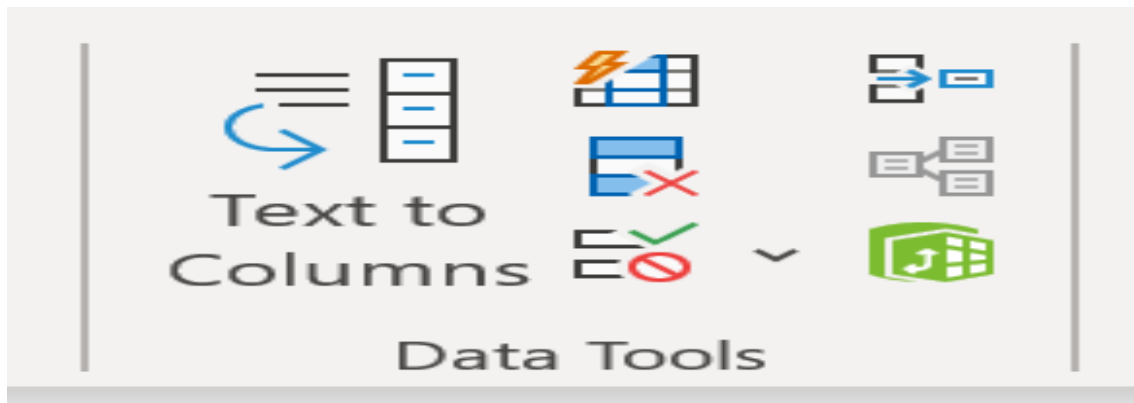


Fig. 1: Data validation tool in Excel

Data Entry Task

The example below illustrates a case of data entry, where the province must be entered for every store location. Since stores are only located in certain provinces, any incorrect entry should be caught.

It is accomplished in Excel using a two-fold data validation. First, the relevant provinces are incorporated into a drop-down menu that allows the user to select from a list of valid provinces.

	A	B	
1	Store ID	Province	
2	5001	AB	
3	5024	ON	
4	5047	BC	
5	5070	AB	
6	5093	SK	
7	5116	NS	
8	5139	NY	
9	5162	BC	
10	5185	AB	
11		ON	

Fig. 2: First level of data validation

Second, if the user inputs a wrong province by mistake, such as “NY” instead of “NS,” the system warns the user of the incorrect input.

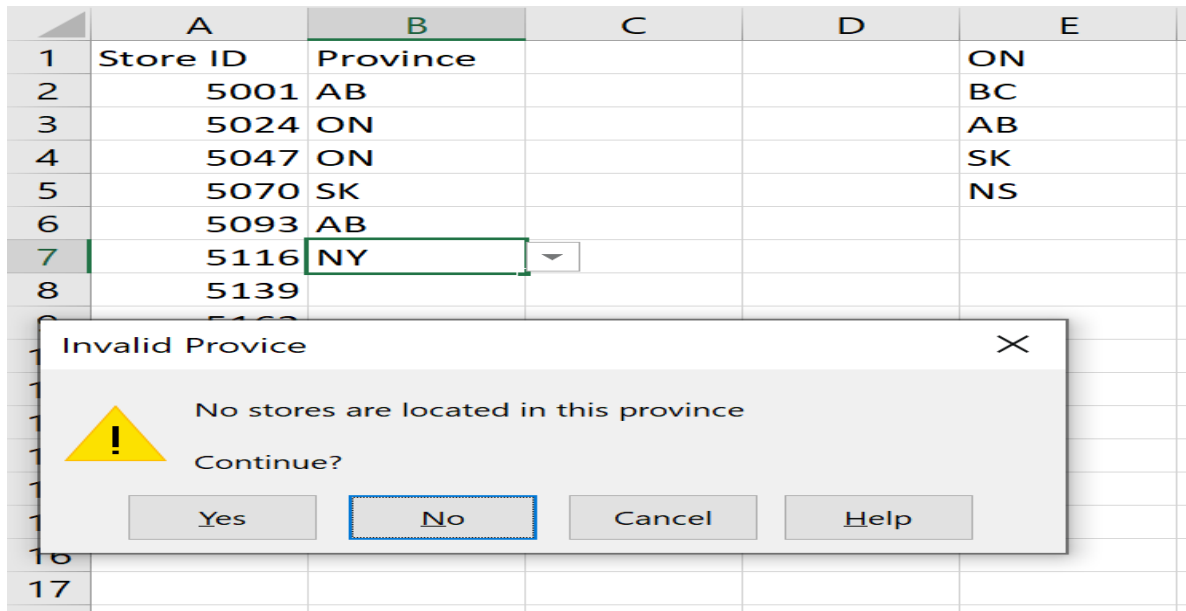


Fig. 3: Second level of data validation

Further, if the user ignores the warning, an analysis can be conducted using the data validation feature in Excel that identifies incorrect inputs.

	A	B	
1	Store ID	Province	
2	5001	AB	
3	5024	ON	
4	5047	ON	
5	5070	SK	
6	5093	AB	
7	5116	NY	
8	5139	BC	
9	5162	AB	
10	5185	ON	
11			

Fig. 4: Final level of data validation

Analysis and Interpretation

What is Data Analysis?

Data analysis is defined as a process of cleaning, transforming, and modeling data to discover useful information for business decision-making. The purpose of Data Analysis is to extract useful information from data and taking the decision based upon the data analysis.

A simple example of Data analysis is whenever we take any decision in our day-to-day life is by thinking about what happened last time or what will happen by choosing that particular decision. This is nothing but analyzing our past or future and making decisions based on it. For that, we gather memories of our past or dreams of our future. So that is nothing but data analysis. Now same thing analyst does for business purposes, is called Data Analysis.

Why Data Analysis?

To grow your business even to grow in your life, sometimes all you need to do is Analysis!

If your business is not growing, then you have to look back and acknowledge your mistakes and make a plan again without repeating those mistakes. And even if your business is growing, then you have to look forward to making the business to grow more. All you need to do is analyze your business data and business processes.

Data Analysis Tools



Data analysis tools make it easier for users to process and manipulate data, analyze the relationships and correlations between data sets, and it also helps to identify patterns and trends for interpretation. Here is a complete list of tools used for data analysis in research.

Types of Data Analysis: Techniques and Methods

There are several **types of Data Analysis** techniques that exist based on business and technology. However, the major Data Analysis methods are:

- Text Analysis
- Statistical Analysis
- Diagnostic Analysis
- Predictive Analysis
- Prescriptive Analysis

Text Analysis

Text Analysis is also referred to as Data Mining. It is one of the methods of data analysis to discover a pattern in large data sets using databases or data mining tools. It used to transform raw data into business information. Business Intelligence tools are present in the market which is used to take strategic business decisions. Overall it offers a way to extract and examine data and deriving patterns and finally interpretation of the data.

Statistical Analysis

Statistical Analysis shows “What happen?” by using past data in the form of dashboards. Statistical Analysis includes collection, Analysis, interpretation, presentation, and modeling of data. It analyses a set of data or a sample of data. There are two categories of this type of Analysis – Descriptive Analysis and Inferential Analysis.

Descriptive Analysis

analyses complete data or a sample of summarized numerical data. It shows mean and deviation for continuous data whereas percentage and frequency for categorical data.

Inferential Analysis

analyses sample from complete data. In this type of Analysis, you can find different conclusions from the same data by selecting different samples.

A good data interpretation process is key to making your data usable. It will help you make sure you’re drawing the correct conclusions and acting on your information.

No matter what, data is everywhere in the modern world. There are two groups and organizations: those drowning in data or not using it appropriately and those benefiting.

Diagnostic Analysis

Diagnostic Analysis shows “Why did it happen?” by finding the cause from the insight found in Statistical Analysis. This Analysis is useful to identify behavior patterns of data. If a new problem arrives in your business process, then you can look into this Analysis to find similar patterns of that problem. And it may have chances to use similar prescriptions for the new problems.

Predictive Analysis

Predictive Analysis shows “what is likely to happen” by using previous data. The simplest data analysis example is like if last year I bought two dresses based on my savings and if this year my salary is increasing double then I can buy four dresses. But of course it’s not easy like this because

you have to think about other circumstances like chances of prices of clothes is increased this year or maybe instead of dresses you want to buy a new bike, or you need to buy a house!

So here, this Analysis makes predictions about future outcomes based on current or past data. Forecasting is just an estimate. Its accuracy is based on how much detailed information you have and how much you dig in it.

Prescriptive Analysis

Prescriptive Analysis combines the insight from all previous Analysis to determine which action to take in a current problem or decision. Most data-driven companies are utilizing Prescriptive Analysis because predictive and descriptive Analysis are not enough to improve data performance. Based on current situations and problems, they analyze the data and make decisions.

Data Analysis Process

The **Data Analysis Process** is nothing but gathering information by using a proper application or tool which allows you to explore the data and find a pattern in it. Based on that information and data, you can make decisions, or you can get ultimate conclusions.

Data Analysis consists of the following phases:

- Data Requirement Gathering
- Data Collection
- Data Cleaning
- Data Analysis
- Data Interpretation
- Data Visualization

Data Requirement Gathering

First of all, you have to think about why do you want to do this data analysis? All you need to find out the purpose or aim of doing the Analysis of data. You have to decide which type of data analysis you wanted to do! In this phase, you have to decide what to analyze and how to measure it, you have to understand why you are investigating and what measures you have to use to do this Analysis.

Data Collection

After requirement gathering, you will get a clear idea about what things you have to measure and what should be your findings. Now it's time to collect your data based on requirements. Once you collect your data, remember that the collected data must be processed or organized for Analysis. As you collected data from various sources, you must have to keep a log with a collection date and source of the data.

Data Cleaning

Now whatever data is collected may not be useful or irrelevant to your aim of Analysis, hence it should be cleaned. The data which is collected may contain duplicate records, white spaces or errors. The data should be cleaned and error free. This phase must be done before Analysis because based on data cleaning, your output of Analysis will be closer to your expected outcome.

Data Analysis

Once the data is collected, cleaned, and processed, it is ready for Analysis. As you manipulate data, you may find you have the exact information you need, or you might need to collect more data. During this phase, you can use data analysis tools and software which will help you to understand, interpret, and derive conclusions based on the requirements.

Data Interpretation

After analyzing your data, it's finally time to interpret your results. You can choose the way to express or communicate your data analysis either you can use simply in words or maybe a table or chart. Then use the results of your data analysis process to decide your best course of action.

Data Visualization

Data visualization is very common in your day to day life; they often appear in the form of charts and graphs. In other words, data shown graphically so that it will be easier for the human brain to understand and process it. Data visualization often used to discover unknown facts and trends. By observing relationships and comparing datasets, you can find a way to find out meaningful information.

What is Data Interpretation

Data interpretation is the process of reviewing data and arriving at relevant conclusions using various analytical research methods. Data analysis assists researchers in categorizing, manipulating, and summarizing data to answer critical questions.

In business terms, the interpretation of data is the execution of various processes. This process analyzes and revises data to gain insights and recognize emerging patterns and behaviors. These conclusions will assist you as a manager in making an informed decision based on numbers while having all of the facts at your disposal.

Importance of Data Interpretation

Raw data is useless unless it's interpreted. Data interpretation is important to businesses and people. The collected data helps make informed decisions.

Make better decisions

Any decision is based on the information that is available at the time. People used to think that many diseases were caused by bad blood, which was one of the four humors. So, the solution was to get rid of the bad blood. We now know that things like viruses, bacteria, and immune responses can cause illness and can act accordingly.

In the same way, when you know how to collect and understand data well, you can make better decisions. You can confidently choose a path for your organization or even your life instead of working with assumptions.

The most important thing is to follow a transparent process to reduce mistakes and tiredness when making decisions.

Find trends and take action

Another practical use of data interpretation is to get ahead of trends before they reach their peak. Some people have made a living by researching industries, spotting trends, and then making big bets on them.

With the proper data interpretations and a little bit of work, you can catch the start of trends and use them to help your business or yourself grow.

Better resource allocation

The last importance of data interpretation we will discuss is the ability to use people, tools, money, etc., more efficiently. For example, If you know via strong data interpretation that a market is underserved, you'll go after it with more energy and win.

In the same way, you may find out that a market you thought was a good fit is actually bad. This could be because the market is too big for your products to serve, there is too much competition, or something else.

No matter what, you can move the resources you need faster and better to get better results.

What are the steps in interpreting data?

Here are some steps to interpreting data correctly.

1. Gather the data

The very first step in data interpretation is gathering all relevant data. You can do this by first visualizing it in a bar, graph, or pie chart. This step aims to analyze the data accurately and without bias. Now is the time to recall how you conducted your research.

Here are two question patterns that will help you to understand better.

- Were there any flaws or changes that occurred during the data collection process?
- Have you saved any observatory notes or indicators?

You can proceed to the next stage when you have all of your data.

2. Develop your discoveries

This is a summary of your findings. Here, you thoroughly examine the data to identify trends, patterns, or behavior. If you are researching a group of people using a sample population, this is the section where you examine behavioral patterns. You can compare these deductions to previous data sets, similar data sets, or general hypotheses in your industry. This step's goal is to compare these deductions before drawing any conclusions.

3. Draw Conclusions

After you've developed your findings from your data sets, you can draw conclusions based on your discovered trends. Your findings should address the questions that prompted your research. If they do not respond, inquire about why; it may produce additional research or questions.

4. Give recommendations

The interpretation procedure of data comes to a close with this stage. Every research conclusion must include a recommendation. As recommendations are a summary of your findings and conclusions, they should be brief. There are only two options for recommendations; you can either recommend a course of action or suggest additional research.

Data interpretation examples

Here are two examples of data interpretations to help you understand it better:

Example 1

Let's say your users fall into four age groups. So a company can see which age group likes their content or product. Based on bar charts or pie charts, they can develop a marketing strategy to reach uninvolved groups or an outreach strategy to grow their core user base.

Example 2

Another example of data analysis is the use of recruitment CRM by businesses. They utilize it to find candidates, track their progress, and manage their entire hiring process to determine how they can better automate their workflow.

Report writing and presentation of results,

Report writing

Reports are usually spread across a vast horizon of topics but are focused on communicating information about a particular topic and a niche target market. The primary motive of research reports is to convey integral details about a study for marketers to consider while designing new strategies.

Certain events, facts, and other information based on incidents need to be relayed to the people in charge, and creating research reports is the most effective communication tool. Ideal research reports are extremely accurate in the offered information with a clear objective and conclusion. These reports should have a clean and structured format to relay information effectively.

What are Research Reports?

Research reports are recorded data prepared by researchers or statisticians after analyzing the information gathered by conducting organized research, typically in the form of surveys or qualitative methods.

A research report is a reliable source to recount details about a conducted research. It is most often considered to be a true testimony of all the work done to garner specificities of research.

The various sections of a research report are:

1. Summary
2. Background/Introduction
3. Implemented Methods
4. Results based on Analysis
5. Deliberation
6. Conclusion

Components of Research Reports

Research is imperative for launching a new product/service or a new feature. The markets today are extremely volatile and competitive due to new entrants every day who may or may not provide effective products. An organization needs to make the right decisions at the right time to be relevant in such a market with updated products that suffice customer demands.

The details of a research report may change with the purpose of research but the main components of a report will remain constant. The research approach of the market researcher also influences the style of writing reports. Here are seven main components of a productive research report:

- **Research Report Summary:** The entire objective along with the overview of research are to be included in a summary which is a couple of paragraphs in length. All the multiple components of the research are explained in brief under the report summary. It should be interesting enough to capture all the key elements of the report.
- **Research Introduction:** There always is a primary goal that the researcher is trying to achieve through a report. In the introduction section, he/she can cover answers related to this goal and establish a thesis which will be included to strive and answer it in detail. This section should answer an integral question: “What is the current situation of the goal?”. After the research design was conducted, did the organization conclude the goal successfully or they are still a work in progress – provide such details in the introduction part of the research report.
- **Research Methodology:** This is the most important section of the report where all the important information lies. The readers can gain data for the topic along with analyzing the quality of provided content and the research can also be approved by other market researchers. Thus, this section needs to be highly informative with each aspect of research discussed in detail. Information needs to be expressed in chronological order according to its priority and importance. Researchers should include references in case they gained information from existing techniques.

- **Research Results:** A short description of the results along with calculations conducted to achieve the goal will form this section of results. Usually, the exposition after data analysis is carried out in the discussion part of the report.
- **Research Discussion:** The results are discussed in extreme detail in this section along with a comparative analysis of reports that could probably exist in the same domain. Any abnormality uncovered during research will be deliberated in the discussion section. While writing research reports, the researcher will have to connect the dots on how the results will be applicable in the real world.
- **Research References and Conclusion:** Conclude all the research findings along with mentioning each and every author, article or any content piece from where references were taken.

Tips for Writing Research Reports

Writing research reports in the manner can lead to all the efforts going down the drain. Here are 15 tips for writing impactful research reports:

- **Prepare the context before starting to write and start from the basics:** This was always taught to us in school – be well-prepared before taking a plunge into new topics. The order of survey questions might not be the ideal or most effective order for writing research reports. The idea is to start with a broader topic and work towards a more specific one and focus on a conclusion or support, which a research should support with the facts. The most difficult thing to do in reporting, without a doubt is to start. Start with the title, the introduction, then document the first discoveries and continue from that. Once the marketers have the information well documented, they can write a general conclusion.
- **Keep the target audience in mind while selecting a format that is clear, logical and obvious to them:** Will the research reports be presented to decision makers or other researchers? What are the general perceptions around that topic? This requires more care and diligence. A researcher will need a significant amount of information to start writing the research report. Be consistent with the wording, the numbering of the annexes and so on. Follow the approved format of the company for the delivery of research reports and demonstrate the integrity of the project with the objectives of the company.
- **Have a clear research objective:** A researcher should read the entire proposal again, and make sure that the data they provide contributes to the objectives that were raised from the beginning. Remember that speculations are for conversations, not for research reports, if a researcher speculates, they directly question their own research.
- **Establish a working model:** Each study must have an internal logic, which will have to be established in the report and in the evidence. The researchers' worst nightmare is to be required to write research reports and realize that key questions were not included.

- **Gather all the information about the research topic.** Who are the competitors of our customers? Talk to other researchers who have studied the subject of research, know the language of the industry. Misuse of the terms can discourage the readers of research reports from reading further.
- **Read aloud while writing.** While reading the report, if the researcher hears something inappropriate, for example, if they stumble over the words when reading them, surely the reader will too. If the researcher can't put an idea in a single sentence, then it is very long and they must change it so that the idea is clear to everyone.
- **Check grammar and spelling.** Without a doubt, good practices help to understand the report. Use verbs in the present tense. Consider using the present tense, which makes the results sound more immediate. Find new words and other ways of saying things. Have fun with the language whenever possible.
- **Discuss only the discoveries that are significant.** If some data are not really significant, do not mention them. Remember that not everything is truly important or essential within research reports.
- **Try and stick to the survey questions.** For example, do not say that the people surveyed "were worried" about an research issue, when there are different degrees of concern.
- **The graphs must be clear enough so that they understand themselves.** Do not let graphs lead the reader to make mistakes: give them a title, include the indications, the size of the sample, and the correct wording of the question.
- **Be clear with messages.** A researcher should always write every section of the report with an accuracy of details and language.
- **Be creative with titles**– Particularly in segmentation studies choose names "that give life to research". Such names can survive for a long time after the initial investigation.
- **Create an effective conclusion:** The conclusion in the research reports is the most difficult to write, but it is an incredible opportunity to excel. Make a precise summary. Sometimes it helps to start the conclusion with something specific, then it describes the most important part of the study, and finally, it provides the implications of the conclusions.
- **Get a couple more pair of eyes to read the report.** Writers have trouble detecting their own mistakes. But they are responsible for what is presented. Ensure it has been approved by colleagues or friends before sending the final draft out.

Presentation of results

Written Presentation of Results

Once you've gone through the process of doing communication research – using a quantitative, qualitative, or critical/rhetorical methodological approach – the final step is to communicate it.

The major style manuals (the APA Manual, the MLA Handbook, and Turabian) are very helpful in documenting the structure of writing a study, and are highly recommended for consultation. But, no matter what style manual you may use, there are some common elements to the structure of an academic communication research paper.

Title Page:

This is simple: Your Paper's Title, Your Name, Your Institutional Affiliation (e.g., University), and the Date, each on separate lines, centered on the page. Try to make your title both descriptive (i.e., it gives the reader an idea what the study is about) and interesting (i.e., it is catchy enough to get one's attention).

For example, the title, "The uncritical idealization of a compensated psychopath character in a popular book series," would not be an inaccurate title for a published study, but it is rather vague and exceedingly boring. That study's author fortunately chose the title, "A boyfriend to die for: Edward Cullen as compensated psychopath in Stephanie Meyer's *Twilight*," which is more precisely descriptive, and much more interesting (Merskin, 2011). The use of the colon in academic titles can help authors accomplish both objectives: a catchy but relevant phrase, followed by a more clear explanation of the article's topic.

Abstract:

In some instances, you might be asked to write an abstract, which is a summary of your paper that can range in length from 75 to 250 words. If it is a published paper, it is useful to include key search terms in this brief description of the paper (the title may already have a few of these terms as well). Although this may be the last thing you write, make it one of the best things you write, because this may be the first thing your audience reads about the paper (and may be the only thing read if it is written badly). Summarize the problem/research question, your methodological approach, your results and conclusions, and the significance of the paper in the abstract.

Quantitative and qualitative studies will most typically use the rest of the section titles noted below. Critical/rhetorical studies will include many of the same steps, but will often have different headings. For example, a critical/rhetorical paper will have an introduction, definition of terms, and literature review, followed by an analysis (often divided into sections by areas of investigation) and ending with a conclusion/implications section. Because critical/rhetorical research is much more descriptive, the subheadings in such a paper are often times not generic subheads like "literature review," but instead descriptive subheadings that apply to the topic at hand, as seen in the schematic below. Because many journals expect the article to follow typical research paper headings of introduction, literature review, methods, results, and discussion, we discuss these sections briefly next.

Social Scientific Project

- Introduction
- Literature Review
- Method
- Results
- Discussion
- References

Critical Projects

- Introduction
- Definition of Terms
- Literature Review
- Analysis (divide sections by areas of investigation)
- Conclusions*

*All sections may not be named this way

Introduction:

As you read social scientific journals (see chapter 1 for examples), you will find that they tend to get into the research question quickly and succinctly. Journal articles from the humanities tradition tend to be more descriptive in the introduction. But, in either case, it is good to begin with some kind of brief anecdote that gets the reader engaged in your work and lets the reader understand why this is an interesting topic. From that point, state your research question, define the problem (see Chapter One) with an overview of what we do and don't know, and finally state what you will do, or what you want to find out. The introduction thus builds the case for your topic, and is the beginning of building your argument, as we noted in chapter 1.

By the end of the Introduction, the reader should know what your topic is, why it is a significant communication topic, and why it is necessary that you investigate it (e.g., it could be there is gap in literature, you will conduct valuable exploratory research, or you will provide a new model for solving some professional or social problem).

Literature Review:

The literature review summarizes and organizes the relevant books, articles, and other research in this area. It sets up both quantitative and qualitative studies, showing the need for the study. For critical/rhetorical research, the literature review often incorporates the description of the historical

context and heuristic vocabulary, with key terms defined in this section of the paper. For more detail on writing a literature review, see Appendix 1.

Methods:

The methods of your paper are the processes that govern your research, where the researcher explains what s/he did to solve the problem. As you have seen throughout this book, in communication studies, there are a number of different types of research methods. For example, in quantitative research, one might conduct surveys, experiments, or content analysis. In qualitative research, one might instead use interviews and observations. Critical/rhetorical studies methods are more about the interpretation of texts or the study of popular culture as communication. In creative communication research, the method may be an interpretive performance studies or filmmaking. Other methods used sometimes alone, or in combination with other methods, include legal research, historical research, and political economy research.

In quantitative and qualitative research papers, the methods will be most likely described according to the APA manual standards. At the very least, the methods will include a description of participants, data collection, and data analysis, with specific details on each of these elements. For example, in an experiment, the researcher will describe the number of participants, the materials used, the design of the experiment, the procedure of the experiment, and what statistics will be used to address the hypotheses/research questions.

Critical/rhetorical researchers rarely have a specific section called "methods," as opposed to quantitative and qualitative researchers, but rather demonstrate the method they use for analysis throughout the writing of their piece.

Helping your reader understand the methods you used for your study is important not only for your own study's credibility, but also for possible replication of your study by other researchers. A good guideline to keep in mind is **transparency**. You want to be as clear as possible in describing the decisions you made in designing your study, gathering and analyzing your data so that the reader can retrace your steps and understand how you came to the conclusions you formed. A research study can be very good, but if it is not clearly described so that others can see how the results were determined or obtained, then the quality of the study and its potential contributions are lost.

Results:

After you completed your study, your findings will be listed in the results section. Particularly in a quantitative study, the results section is for revisiting your hypotheses and reporting whether or not your results supported them, and the statistical significance of the results. Whether your study supported or contradicted your hypotheses, it's always helpful to fully report what your results

were. The researcher usually organizes the results of his/her results section by research question or hypothesis, stating the results for each one, using statistics to show how the research question or hypothesis was answered in the study.

The qualitative results section also may be organized by research question, but usually is organized by themes which emerged from the data collected. The researcher provides rich details from her/his observations and interviews, with detailed quotations provided to illustrate the themes identified. Sometimes the results section is combined with the discussion section.

Critical/rhetorical researchers would include their analysis often with different subheadings in what would be considered a "results" section, yet not labeled specifically this way.

Discussion:

In the discussion section, the researcher gives an appraisal of the results. Here is where the researcher considers the results, particularly in light of the literature review, and explains what the findings mean. If the results confirmed or corresponded with the findings of other literature, then that should be stated. If the results didn't support the findings of previous studies, then the researcher should develop an explanation of why the study turned out this way. Sometimes, this section is called a "conclusion" by researchers.

References:

In this section, all of the literature cited in the text should have full references in alphabetical order. Appendices: Appendix material includes items like questionnaires used in the study, photographs, documents, etc. An alphabetical letter is assigned for each piece (e.g. Appendix A, Appendix B), with a second line of title describing what the appendix contains (e.g. Participant Informed Consent, or New York Times Speech Coverage). They should be organized consistently with the order in which they are referenced in the text of the paper. The page numbers for appendices are consecutive with the paper and reference list.

Tables/Figures:

Tables and figures are referenced in the text, but included at the end of the study and numbered consecutively. (Check with your professor; some like to have tables and figures inserted within the paper's main text.) Tables generally are data in a table format, whereas figures are diagrams (such as a pie chart) and drawings (such as a flow chart).

Importance of report writing

- 1. Decision Making Tool:** Today's complex business organizations require thousands of information. Reports provide the required information. A large number of important decisions in business or any other areas are taken on the basis of information presented in the report.
- 2. Investigation:** Whenever there is any problem, a committee or commission or study group investigate the problem to find out the reason behind the problem and present the found output with or without the recommendation in the form of a report.
- 3. Evaluation:** Large scale organization are engaged in multidimensional activities. It is not possible for a single top executive to keep personal watch on what others are doing. So, the executive depends on reports to evaluate the performance of various departments or units.
- 4. Quick Location:** There is no denying the fact that business executives need information for quick decision making. As top executive are found to be busy for various purposes, they need vital sources of information. Such sources can be business report.
- 5. Development of Skill:** Report writing skill develops the power of designing, Organization co-ordination, judgement and communication. Therefore such skill acts as a catalyst.
- 6. Neutral Presentation of Facts:** Facts are required to be presented in a neutral way. Such presentation is ensured through a report as it investigates, explains and evaluates and much more about any fact independently.
- 7. Professional Advancement:** Report also plays a major role in professional progress and advancement.
For promotion to the rank and file position, satisfactory job performance is enough to help a person. But for promotion to high level position, intellectual ability is highly required. Such ability can be expressed through the report submitted to higher authority.
- 8. Proper Control:** Whether activities are happening according to plan or not is expressed through a report. So, controlling such activities are implemented based on the information of a report.
- 9. A Managerial Tool:** Various reports make activities easy for the managers. For planning, organizing, coordinating, motivating and controlling, A manager may find help from a report which acts as a source of information.
- 10. Encountering Advance and Complex Situation:** In a large business organization, there is always some sort of labor problems which may bring complex situations. To tackle those situation, managers take the help of a report.

Types of research reports

1. Thesis

Thesis is a type of research report. A thesis is a long-form research document that presents the findings and conclusions of an original research study conducted by a student as part of a graduate or postgraduate program. It is typically written by a student pursuing a

higher degree, such as a Master's or Doctoral degree, although it can also be written by researchers or scholars in other fields.

2. Research Paper

Research paper is a type of research report. A research paper is a document that presents the results of a research study or investigation. Research papers can be written in a variety of fields, including science, social science, humanities, and business. They typically follow a standard format that includes an introduction, literature review, methodology, results, discussion, and conclusion sections.

3. Technical Report

A technical report is a detailed report that provides information about a specific technical or scientific problem or project. Technical reports are often used in engineering, science, and other technical fields to document research and development work.

4. Progress Report

A progress report provides an update on the progress of a research project or program over a specific period of time. Progress reports are typically used to communicate the status of a project to stakeholders, funders, or project managers.

5. Feasibility Report

A feasibility report assesses the feasibility of a proposed project or plan, providing an analysis of the potential risks, benefits, and costs associated with the project. Feasibility reports are often used in business, engineering, and other fields to determine the viability of a project before it is undertaken.

6. Field Report

A field report documents observations and findings from fieldwork, which is research conducted in the natural environment or setting. Field reports are often used in anthropology, ecology, and other social and natural sciences.

7. Experimental Report

An experimental report documents the results of a scientific experiment, including the hypothesis, methods, results, and conclusions. Experimental

reports are often used in biology, chemistry, and other sciences to communicate the results of laboratory experiments.

8. Case Study Report

A case study report provides an in-depth analysis of a specific case or situation, often used in psychology, social work, and other fields to document and understand complex cases or phenomena.

9. Literature Review Report

A literature review report synthesizes and summarizes existing research on a specific topic, providing an overview of the current state of knowledge on the subject. Literature review reports are often used in social sciences, education, and other fields to identify gaps in the literature and guide future research.

Report structure

(A)-Front Matter

- 1) Letter of transmittal (Cover letter)
- 2) Title page
- 3) Executive summary
- 4) Table of content
- 5) List of figures/ Illustration

(B)-Body of the report

- 6) Introduction
- 7) Findings and Documentation
- 8) Conclusion and Recommendation

(C)-Back Matter

- 9) Appendices, References, and Glossary

Note: The letter of transmittal is an optional element of reporting which may or may not be included depending on the type of report prepared.

(A)Front Matter

The front matter consists of all the opening components of a business report. This includes a title page, executive summary, table of contents, and a list of figures/illustrations. A letter of transmittal, which serves as an official cover letter, may also be included.

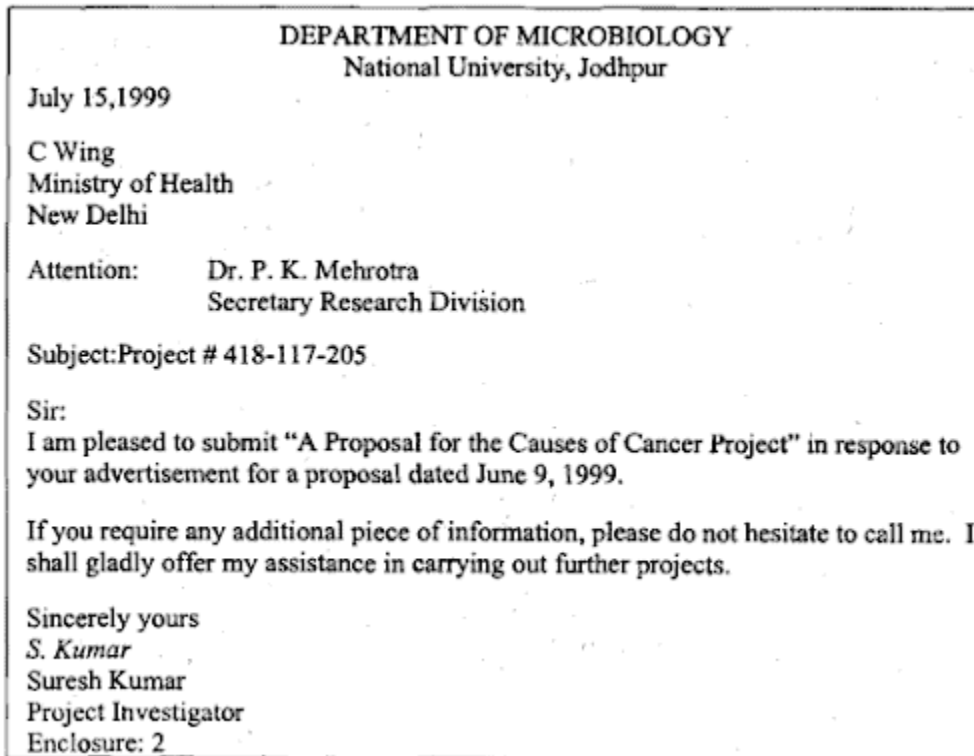
The following elements are included in the front matter of a report:

1) Letter of Transmittal (Cover):

The letter of transmittal serves as a customary cover page. Its purpose is to state the title of the report and if necessary, briefly preview conclusions, state the purpose, and mention key highlights.

It mentions who commissioned the report and appreciates the efforts of those involved in the project. This letter is attached to the top of the report and is written in a courteous tone.

An Example of the Letter of Transmittal



An example of the letter of transmittal.

2) Title Page:

As the name suggests, the title page is the opening page of a business report. It comprises the following:

1. The title of the subject
2. Name of the writer
3. Name of the receiver, title, and organization
4. Purpose of the report
5. Date of submission

The title of a business report should clearly convey the type of report and its purpose. It should be written in capital letters and centered in alignment.

An Example of the Title Page

REMOTE SENSING IN INDIA; A STUDY A Paper Presented to Prof. U. C. Nag Computer Centre, Meerut Open University by Vinay Kumar In Partial Fulfilment Of the Requirements for the Postgraduate Diploma in Remote Sensing On 23 March 1999

3) Executive Summary:

The executive summary or the abstract is a major element of a business report. It is written for busy executives who may only read this part of the report. It offers a summary of the entire report and its findings.

It includes:

1. Purpose of the report
2. Scope of the investigation
3. The methods used to conduct the investigation
4. Key findings along with essential statistics
5. Conclusions, analysis, and recommendations

[R&D] Q2 explorations into new watch series — Executive Summary

Problem statement

In recent customer feedback sessions, 52% of customers have expressed a need for a simpler and cheaper version of our product. In surveys of customers who have chosen competitor watches, price is mentioned 87% of the time. To best serve our existing customers, and to branch into new markets, we need to develop a series of watches that we can sell at an appropriate price point for this market.

Proposed solution

Our new watch series will begin at 20% cheaper than our current cheapest option, with the potential for 40%+ cheaper options depending on material and movement. In order to offer these prices, we will do the following:

- Offer watches in new materials, including potentially silicone or wood
- Use high-quality quartz movement instead of in-house automatic movement
- Introduce customizable band options, with a focus on choice and flexibility over traditional luxury

Note that every watch will still be rigorously quality controlled in order to maintain the same world-class speed and precision of our current offerings.

Value

With new offerings that are between 20% and 40% cheaper than our current cheapest option, we expect to be able to break into the casual watch market, while still supporting our luxury brand. That will help us hit FY22's Objective 3: Expanding the brand. These new offerings have the potential to bring in upwards of three million dollars in profits annually, which will help us hit FY22's Objective 1: 7 million dollars in annual profit.

Early customer feedback sessions indicate that cheaper options will not impact the value or prestige of the luxury brand, though this is a risk that should be factored in during design. In order to mitigate that risk, the product marketing team will begin working on their go-to-market strategy six months before the launch.

Final thoughts & next steps

Cheaper and varied offerings not only allow us to break into a new market—it will also expand our brand in a positive way. With the attention from these new offerings, plus the anticipated demand for cheaper watches, we expect to increase market share by 2% annually. For more information, read our [go-to-market strategy](#) and [customer feedback documentation](#).

Source: asana.com

Note: You can further check out the following blog by Hubspots on powerfull executive summary for more details.

4) Table of Contents:

For longer and more detailed information, a table of contents is a necessary element of report writing. It provides the readers with an overview of the content present. This helps them easily locate particular sections of a report. All headings and subheadings are listed along with page numbers in the exact order of appearance.



INTEGRATED ANNUAL REPORT 2020-21	
Contents	
OVERVIEW	
02	At a Glance
04	Chairman's Statement
06	Board of Directors
07	Management Committee
08	Our Performance
09	Financial Performance
10	India Sustainability Initiatives (Highlights)
12	Our Strategy
16	Our Business Model
REPORTS	
18	Report of Board of Directors and Management Discussion and Analysis
56	Annual Report on Corporate Social Responsibility
66	Business Responsibility Report
80	Corporate Governance Report
107	Secretarial Audit Report
110	Economic Value Added
FINANCIAL STATEMENTS	
Standalone Financial Statements	
112	Independent Auditors' Report
124	Balance Sheet
126	Statement of Profit and Loss
128	Statement of Changes in Equity
130	Statement of Cash Flows
132	Notes
Consolidated Financial Statements	
192	Independent Auditors' Report
202	Balance Sheet
204	Statement of Profit and Loss
206	Statement of Changes in Equity
208	Statement of Cash Flows
210	Notes
269	Form AOC-1
OTHERS	
270	Awards & Recognition
271	Corporate Information
272	Notice of Annual General Meeting
282	Profile of Directors

5) List of Figures/Illustrations:

The graphic aids (charts, graphs, photographs, diagrams, etc) used in a business report are organized in their own separate table of contents. A list of figures or list of illustrations helps the readers locate the specific page where this information is presented.

If a report has only tables, it is called a list of tables. If it has only figures with no tables, it is termed a list of figures. And if it contains both tables and figures, it is collectively called a list of illustrations.

An Example of a List of Illustrations		
LIST OF ILLUSTRATIONS		
Figures		
Figure 1	Compilation-execution of FORTRAN program	3
Figure 2	Example of EXEC statement to invoke compiler	6
Figure 3	Example of compilation feature	8
Figure 4	Text of FORTXCL catalogued procedure	9
Figure 5	Structure of segment	14
Figure 6	Composition of error-control table	21
Tables		
Table 1	Job control statements	2
Table 2	Types of compiler options	4
Table 3	Contents of object list	7
Table 4	Return codes of loader	10
Table 5	ANS control characters for line feed control	11
Table 6	Standard values of error-control table	16
(Source, Markel 1984: 247)		

(B)Body of a Report

The body contains the meaty content of the report. After the introductory sections, is where the substance of the report lies. The body of a report may vary according to the type of business report. For example, The essential elements of a technical report will differ from the content of a progress report.

The following sections make up the body of a report:

6) Introduction:

The introduction is the ‘beginning’ of the report. This is the section where the structure of the report is detailed. Along with this, key questions are answered such as “what is the report about ” and “how can it be used”.

This is where the page numbering of a report starts. The introduction of a report is used to:

- Define the main subject or problem
- Provide brief background information
- State the overall purpose of the report
- State theoretical foundations, scope, methodology, and any limitations

7) Findings and Documentation:

This is one of the main elements of a report. It is also usually the longest section of a report. The factual findings are presented along with an interpretation of the data collected.

The key findings of the investigation are explained in detail. The writer mentions all the sources used to formulate the report. This helps in the authentication of the report and allows the readers to visit the source material for further information.

All significant information gathered is documented in this section. Both positive and negative points are mentioned to form conclusions and recommend solutions.

The relevant questions pertaining to the report, its purpose, addressing problems and analysis of information are present in the findings and documentation.

8) Conclusion:

The findings and purpose of the report are summarized in the conclusion. The information presented in the report is analyzed and the advantages, limitations, and actions that need to be taken are stated in this section.

The writer must avoid including new information in the conclusion. It must only be relevant to the data presented in the rest of the report.

9) Recommendations:

The recommendation is one of the prime elements of the structure of a business report, this part includes the recommended actions that need to be taken based on the findings of the report. They are relevant to:

- What must be done
- Who will do it
- How and when they will do it
- Where it will be done

Recommendations can be made for improvement, implementation of new ideas, or change which will help the company's performance.

4.1 Suggestions

- In order to have interactive sessions, teacher should have an interaction with student even through informal channel when needed in order to keep sense of belonging. The informal channel may include (instant messages online chat groups, audio calls, private video calls ...) with equal importance given to formal channels such as (online platforms, email ...).
- At the end students and teacher should collaborate and work on developing different kind of activates and incentives in order to reduce the boredom and increase participation in class.

4.2 Conclusions

- Pandemic had turned out to be a disaster for the planet, providing chance and opportunity to restructure current ongoing educational system dependent on classrooms. Though there have been some major challenges such as creating a connection within, teaching, curriculum assessment, and learning in school and non-school environments to meet the 'new normal'.
- Learn ways to empower students and teachers and parents to manage new challenges in 'new normal' schooling. On the other hand, the sudden changes due to external factor in online education has given advantage to students for picking up their skill areas and joining the online education platform to enhance their skill.

(C)Back Matter

Following the front matter and the body of the report, we have the back matter. This section of the report highlights auxiliary and supporting segments such as glossary, references, and appendices.

10) Glossary:

The glossary serves as a dictionary providing definitions of technical terms used in the report. The words are listed in alphabetical order.

The glossary is listed before the appendix and if very brief, it can be placed just after the table of contents.

11) References:

References are also one of the essential components of a business report because a report is only considered authentic if all accurate sources are listed. All references from where the data was collected are mentioned in this section. Data collected from third-party sources such as stats, facts, and journals must be credited to their original sources.

References-

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12) Appendices:

The appendix is the last element of the business report structure that contains supporting information that would be too bulky to include in the body of the report. This is considered an optional section that may or may not be included in a report.

It contains specialized information such as maps, questionnaires, interviews, letters, etc.

Guidelines for effective documentation

Rule 1. Never consider the time you invest in writing or planning documentation as wasted time

Like any well-designed project, the production of good documentation requires clear planning and a considerable investment of time and resources. This may seem like a pain to you, especially when you are overloaded with other work. Rather than viewing documentation and the lengthy meetings with the "non-specialised" technical writer who needs to be explained step by step the simplest things, as a necessary evil, you should see it as a tool for helping you clarify your product to yourself as well as to others.

One of the problems of design is that when you become bogged down in the gritty details and mechanics you may lose sight of the forest, for all the trees. Documentation not only provides you with a roadmap of the forest, it also presents an overall perspective, an opportunity to step out of the logger's mindset and see the larger plan or goal. Where are you going? Are you missing anything? Have you included everything that needs to be included?

Your discussions with your writers may help identify problems that you were not aware. The documentation process helps you to place what you are doing in perspective and keep in focus the ultimate purpose or use of the product being designed.

Preferably an outline should be produced that covers the major sections of the documentation. The outline is designed to lead the reader in a logical manner through the task, procedure or description being described.

Make sure your time allotment for a project takes into consideration time devoted to documentation.

Rule 2. Don't take what you say or write for granted

What may seem clear and obvious to you may not appear so to someone from outside the company, who is not familiar with the concepts and jargon that you use daily.

When a writer asks you what you mean by something, the answer may seem obvious, but it may not be. Language needs to be taken apart, examined for its basic concepts and then reassembled. Often, drafts need to be rewritten several times, to clarify the meaning and focus the text to fit the reader's needs.

Information that has been assembled from a technical or development perspective needs to be presented in a form suitable for the purpose of the document and the intended reader. Readers of manuals or brochures will probably not need to know any technical details, other than on a broad, conceptual level. This requires removing information not relevant to the reader.

Rule 3. Provide plenty of feedback and input

The basic formula for documentation is:

Quality of input = quality and clarity of output

Remember this formula the next time you give feedback. Writers depend on receiving good quality information. You are responsible for ensuring that the information you provide is comprehensive and accurate.

Feedback should be concise and to the point. It should:

- Refer to specific items
- Provide clear guidelines for what needs to be changed
- Preferably be presented as a list of to do items or as text to be edited/changed in the document

You may not be sure of the precise problem with the documentation. Something may feel wrong or be missing, or a description of a concept or procedure may feel incomplete. Trust your gut feeling and explore this further. You know the product. Ask yourself or the writer “What do you mean by this?” Try rephrasing unclear sections in a way that communicates the way that you see the product.

Use diagrams and examples to explain what you mean.

Finally, if a writer requests information or feedback, don’t procrastinate and delay. This holds up the entire documentation process, since the writer cannot proceed until your feedback or information has been received.

Rule 4. Be prepared for several drafts

Clear documentation, like any other activity, requires constant rethinking and reshaping. Concepts, terminology and processes need to be identified, written about, clarified, rewritten about, discussed and honed. Writing is a cyclical process. Each round of feedback produces a document that is clearer to read, more comprehensive and accurate. The more input received from different perspectives and different readers, the more reliable and accurate will be the nature of the information you produce.

Unfortunately, the myth of dashing off a draft in one sitting, which contains everything that needs to be said in a clear way, is impractical. It would be similar to asking a software engineer to produce a program in one draft.

Rule 5. Be on friendly terms with your writer

The writer assigned to your project is there to help you. He/she is not an added nuisance that is a drain on your resources and time. The writer’s purpose is to take the problems and considerations

that go with documentation off your hands.

The task of documenting a project is both lengthy and time- consuming – technical writers are employed to take over responsibility, allowing you to devote your energies where they are most needed.

Don't cut short or avoid your writer, for fear of wasting your time. Your patience will pay off.

Rule 6. Make sure all stages of design are well-documented

It pays to invest time and energy at the start of a project in clarifying the concepts of your design, through white papers, product proposals, presentations, diagrams and system specification documents. The more complete the documentation, the more information will be available at a later stage for user-orientated manuals and other external documents.

Rule 7. Involve your writer in all stages of the product design

Your writer is a valuable resource. Don't involve him/her at the last stage, in the final rush to complete the project.

The writer should be involved in all stages of a project. An experienced writer will be able to provide a bird's eye view of the entire project and its components. Such a writer can integrate and synthesise information from different sources, written over different time periods. A writer who has been involved in all stages of documentation will be able to fill in gaps and make critical decisions about the contents of a manual, with the minimum of investment from the product manager and engineers.

Rule 8. Writing is team work

The final documentation product should be the result of the combined input and feedback from the entire team: engineers, product managers, writers and quality assurance staff. Each professional member of the team can add something from his or her own perspective that can contribute to the value of the documentation and the project. Remember this when you are asked to offer feedback or if it is your task to coordinate and ensure that other members of the team contribute their feedback.

Rule 9. Be Flexible in your expectations and demands

Be prepared to be flexible in your expectations and demands. You may have to share a writer with several other product managers, each with their own deadlines. A busy writer, with several projects on hand, may not be in a position to drop everything else and devote all their time to your project. It is therefore important that you plan your documentation needs in advance, together with the writer. This will enable him/her to schedule time for you.
